14.7 FRESNEL'S TWO-MIRROR ARRANGEMENT LO 6

After Young's double hole interference experiment, Fresnel devised a series of arrangements to produce the interference pattern. One of the experimental arrangements, known as the **Fresnel two-mirror arrangement**, is shown in Fig. 14.18; it consists of two plane mirrors which are inclined to each other at a small angle θ and touching at the point M. S represents a narrow slit placed perpendicular to the plane of the paper.

A portion of the wavefront from S gets reflected from M_1M and illuminates the region AD of the screen. Another portion of the wavefront gets reflected from the mirror MM_2 and illuminates the region BC of the screen. Since these two wavefronts are derived from the same source they are coherent. Thus in the region BC, one observes interference fringes. The formation of the fringes can also be understood as being due to the interference of the wavefronts from the virtual sources S_1 and S_2 of S formed by the mirrors M_1 and M_2, respectively. From simple geometric considerations, it can be shown that the points S, S_1 and S_2 lie on a circle whose centre is at the point M. Further, if the angle between the mirrors is θ, then the angle S_1SS_2 is also θ and the angle S_1MS_2 is 2θ. Thus, S_1S_2 is $2R\theta$, where R is the radius of the circle.

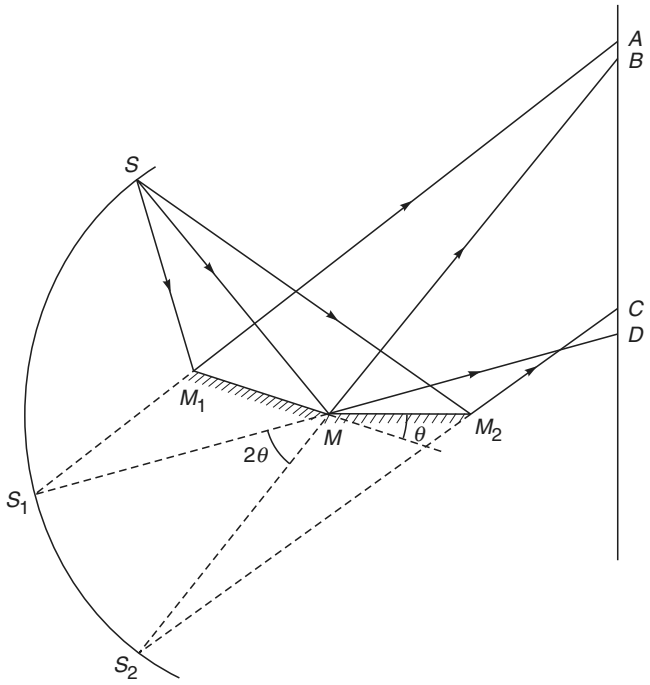


Fig. 14.18 Fresnel's two-mirror arrangement.

14.8 FRESNEL BIPRISM LO 6

Fresnel devised yet another simple arrangement for the production of interference pattern. He used a biprism, which was actually a simple prism, the base angles of which are extremely small (~ 20′). The base of the prism is shown in Fig. 14.19 and the prism is assumed to stand perpendicular to the plane of the paper. S represents the slit which is also placed perpendicular to the plane of the paper. Light from the slit S gets refracted by the prism and produces two virtual images S_1 and S_2. These images act as coherent sources and produce interference fringes on the right of the biprism. The fringes can be viewed through an eyepiece. If n represents the refractive index of the material of the biprism and α the base angle, then $(n-1)\alpha$ is approximately the angular deviation produced by the prism and, therefore, the distance S_1S_2 is $2a(n-1)\alpha$, where a represents the distance from S to the base of the prism. Thus, for $n = 1.5$, $\alpha \approx (20') \approx 5.8 \times 10^{-3}$ radians, $a \approx 2$ cm, one gets $d = 0.012$ cm.

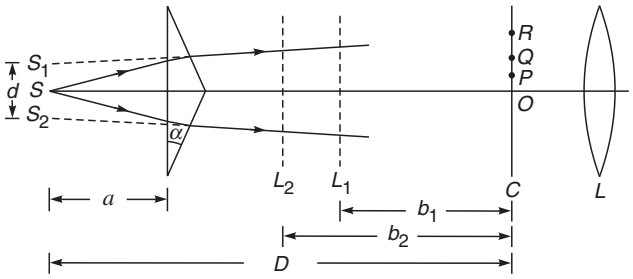


Fig. 14.19 Fresnel's biprism arrangement. C and L represent the positions of the crosswires and the eyepiece, respectively. In order to determine d one introduces a lens between the biprism and the crosswires; L_1 and L_2 represent the two positions of the lens where the slits are clearly seen.

The biprism arrangement can be used for the determination of wavelength of an almost monochromatic light like the one coming from a sodium lamp. Light from the sodium lamp illuminates the slit S and interference fringes can be easily viewed through the eyepiece. The fringe width (β) can be determined by means of a micrometer attached to the eyepiece. Once β is known, λ can be determined by using the following relation:

$$\lambda = \frac{d\beta}{D} \qquad \textbf{(14.34)}$$

It may be mentioned that in order to determine d, one need not measure the value of α. In fact the distances d and D can easily be determined by placing a convex lens between the

biprism and the eyepiece. For a fixed position of the eyepiece there will be two positions of the lens (shown as L_1 and L_2 in Fig. 14.19) where the images of S_1 and S_2 can be seen at the eyepiece.* Let d_1 be the distance between the two images when the lens is at the position L_1 (at a distance b_1 from the eyepiece). Let d_2 and b_2 be the corresponding distances when the lens is at L_2. Then, it can easily be shown that

$$d = \sqrt{d_1 d_2}$$

and

$$D = b_1 + b_2$$

Typically for $d \approx 0.01$ cm, $\lambda \approx 6 \times 10^{-5}$ cm, $D \approx 50$ cm, $\beta \approx 0.3$ cm.

In the above arrangement we have considered a slit instead of a point source. Since each pair of points S_1 and S_2 produce (approximately) straight line fringes, the slit will also produce straight line fringes of increased intensity.

14.9 INTERFERENCE WITH WHITE LIGHT

LO 7

We will now discuss the interference pattern when the slit is illuminated by white light. The wavelengths corresponding to the violet and red end of the spectrum are about 4×10^{-5} cm and 7×10^{-5} cm, respectively. Clearly, the central fringe produced at the point O (Fig. 14.19) will be white because all wavelengths will constructively interfere here. Now, slightly below (or above) the point O the fringes will become colored. For example, if the point P is such that

$$S_2P \sim S_1P = 2 \times 10^{-5} \text{ cm} \left(= \frac{\lambda_{\text{violet}}}{2} \right)$$

then complete destructive interference will occur only for the violet color. Partial destructive interference will occur for other wavelengths. Consequently, we will have a line devoid of the violet color and will appear reddish. The point Q which satisfies

$$S_2Q \sim S_1Q = 3.5 \times 10^{-5} \text{ cm} \left(= \frac{\lambda_{\text{red}}}{2} \right)$$

will be devoid of the red color. It will correspond to almost constructive interference for the violet color. No other wavelength (in the visible region) will neither constructively nor destructively interfere. Thus following the white central fringe we will have colored fringes; when the path difference is about 2×10^{-5} cm the fringe will be red in color, then the color will gradually change to violet. The colored fringes will soon disappear because at points far away from O there will be so many wavelengths (in the visible region) which will constructively interfere that we will observe uniform white illumination. For example, at a point R, such that $S_2R \sim S_1R = 30 \times 10^{-5}$ cm, wavelengths corresponding to $30 \times 10^{-5}/n$ ($n = 1, 2, \ldots$) will constructively interfere. In the visible region, these wavelengths will be 7.5×10^{-5} cm (red), 6×10^{-5} cm (yellow), 5×10^{-5} cm (greenish yellow) and 4.3×10^{-5} cm (violet). Further, wavelengths corresponding to $30 \times 10^{-5}/\left(n + \frac{1}{2}\right)$ will destructively interfere; thus, in the visible region, the wavelengths 6.67×10^{-5} cm (orange), 5.5×10^{-5} cm (yellow), 4.6×10^{-5} cm (indigo) and 4.0×10^{-5} cm (violet) will be absent. The color of such light, as seen by the unaided eye, will be white. Thus, with white light one gets a white central fringe at the point of zero path difference along with a few colored fringes on both the sides, the color soon fading off to white. While using a white light source, if we put a red (or green) filter in front of our eye, we will see the interference pattern corresponding to the red (or green) light.

In the usual interference pattern with a nearly monochromatic source (like a sodium lamp) a large number of interference fringes are obtained and it is extremely difficult to determine the position of the central fringe. In many interference experiments, it is necessary to determine the position of the central fringe and, as has been discussed above, this can easily be done by using white light as a source.

As discussed above, when we observe interference pattern using a white light source, we will see only few colored fringes. However, if we put a red filter in front of our eye, the fringe pattern (corresponding to the red color) will suddenly appear. If we replace the red filter by a green filter in front of our eye, the fringe pattern corresponding to the green color will appear.

14.10 DISPLACEMENT OF FRINGES

LO 8

We will now discuss the change in the interference pattern produced by introducing a thin transparent plate in the path of one of the two interference beams as shown in Fig. 14.20. Let t be the thickness of the plate and let n be its refractive

* This method is similar to the displacement method for the determination of the focal length of a convex lens.